

Technical Proposal — Extinction 101

1. Problem Understanding

Our objective is to predict whether an asteroid will result in a **fatal impact** based on its observable characteristics. The problem is a highly imbalanced classification task, where fatal impacts represent a small minority of observations. Therefore, a useful model should be able to identify the rare fatal cases rather than simply predicting the majority class.

2. Data Processing & Feature Engineering

We first inspected the dataset for missing values, invalid entries, duplicates and inconsistent data types before performing the necessary preprocessing. We removed **asteroid_id** as it is an identifier rather than a meaningful predictive feature. We also excluded **crater_size_km**, as this information would only be available after an impact and could introduce data leakage. To better represent potentially important relationships, we created features such as **log-transformed asteroid diameter** and an **impact-energy proxy** based on asteroid size and velocity.

3. Model Selection & Justification

We established a simple baseline model and compared it with tree-based classification approaches. Tree-based models were considered because they can capture non-linear relationships between asteroid characteristics. Since fatal impacts are relatively rare, we also considered **class weighting or oversampling** to improve learning of the minority class, while ensuring these techniques were applied only to the training data. The final approach was selected based on validation performance and consistency rather than complexity alone.

4. Evaluation Strategy

We used a **stratified train-validation split** to preserve the proportion of fatal and non-fatal cases. We evaluated the model using measures appropriate for an imbalanced classification problem and examined its ability to identify fatal impacts while considering false positives and false negatives. A separate test set was retained for final evaluation.

5. Limitations & Risks

The limited number of fatal examples may make validation results less stable and increase the risk of overfitting. The model may also struggle with asteroid characteristics that differ substantially from those seen during training. With additional data, we would explore richer features, more robust validation and alternative modelling approaches.